Problem · Unit 3 · 10 points · 60 min · Standard exam level

P16 — Particle on a free wedge: a moving constraint, the reaction, momentum and energy, and the limit of a massless wedge

*A wedge of mass M rests on a smooth horizontal table. Its upper face is a smooth plane inclined at the angle α ($0 < \alpha < \pi/2$) to the horizontal, descending towards $-x$; the top edge is A and the corner of the wedge under A is B. A particle P of mass m is placed on the incline. The inertial frame is $S_0 = \{O; \mathbf{i}, \mathbf{j}\}$ (x horizontal, y vertical upwards, gravity $-g\,\mathbf{j}$). The wedge translates without rotating (it does not tip), and its position is the abscissa X of B. The position of P on the incline is the distance $s = |AP|$ measured from A **down the slope**, along the unit vector $\mathbf{u} = -\cos\alpha\,\mathbf{i} - \sin\alpha\,\mathbf{j}$; $\mathbf{n} = -\sin\alpha\,\mathbf{i} + \cos\alpha\,\mathbf{j}$ is the unit normal of the incline pointing out of the wedge. Both bodies are released from rest at $t = 0$ with $s = 0$.*

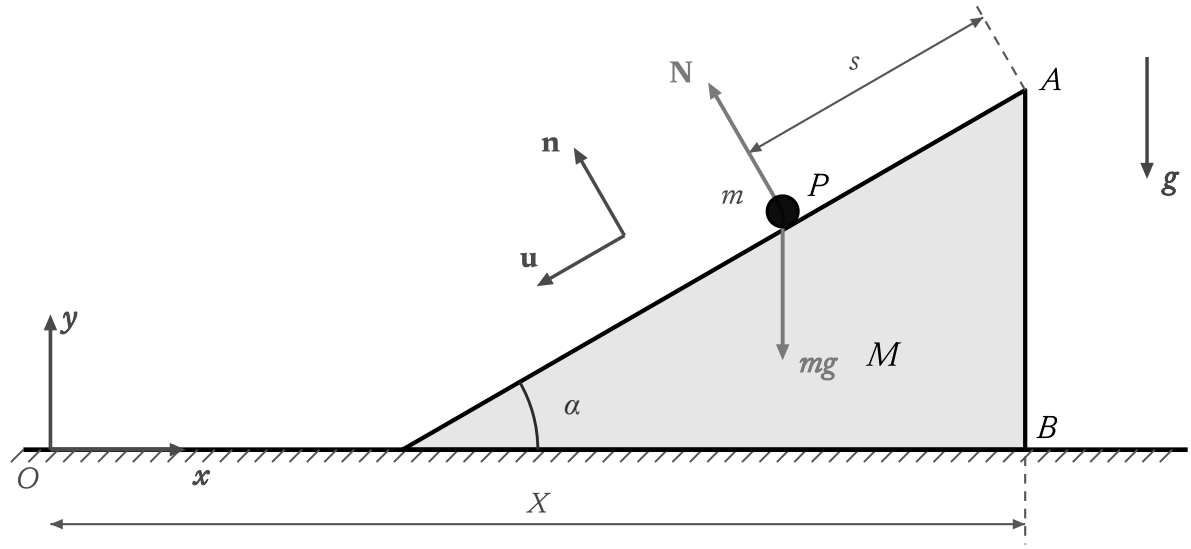


Figure 1. Free wedge: the wedge (mass M) slides on the smooth table, its position is X (abscissa of the corner B); P (mass m) slides on the smooth incline of angle α, at the distance s from the top edge A measured down the slope (unit vector $\mathbf{u}$); $\mathbf{N} = N\mathbf{n}$ is the normal reaction on P (outward normal $\mathbf{n}$).

Setup

- Data: m, M, α, g. Numerical parts: $\alpha = 30°$, $M = 2m$, and the particle has slid a distance L along the incline.
- Hypotheses: point particle, rigid wedge translating on the table, all contacts smooth (no friction), S_0 inertial; the contact of P with the incline is unilateral ($N \geq 0$, checked in (d)).

- Frames: S_0 and the frame $S_1 = \{A; \mathbf{n}, \mathbf{u}\}$ attached to the wedge, ordered so that it is right-handed: $\mathbf{n} \times \mathbf{u} = \big[(-\sin\alpha)(-\sin\alpha) - \cos\alpha\,(-\cos\alpha)\big]\mathbf{k} = \mathbf{k}$ (the order $\{\mathbf{u}, \mathbf{n}\}$ would give $\mathbf{u} \times \mathbf{n} = -\mathbf{k}$, a left-handed basis). S_1 **translates** with acceleration $\ddot{X}\,\mathbf{i}$ and does not rotate.
- Basis change: $\mathbf{i} = -\sin\alpha\,\mathbf{n} - \cos\alpha\,\mathbf{u},\ \mathbf{j} = \cos\alpha\,\mathbf{n} - \sin\alpha\,\mathbf{u}$ (from $\mathbf{i}\cdot\mathbf{n} = -\sin\alpha, \mathbf{i}\cdot\mathbf{u} = -\cos\alpha$, $\mathbf{j}\cdot\mathbf{n} = \cos\alpha, \mathbf{j}\cdot\mathbf{u} = -\sin\alpha$).

In matrix form:

$$\begin{pmatrix}\mathbf{i}\\ \mathbf{j}\end{pmatrix} = [R]\begin{pmatrix}\mathbf{n}\\ \mathbf{u}\end{pmatrix}, \qquad [R] = \begin{pmatrix}-\sin\alpha & -\cos\alpha\\ \cos\alpha & -\sin\alpha\end{pmatrix}$$

with $[R][R]^T = \begin{pmatrix}\sin^2\alpha + \cos^2\alpha & -\sin\alpha\cos\alpha + \cos\alpha\sin\alpha\\ -\sin\alpha\cos\alpha + \cos\alpha\sin\alpha & \cos^2\alpha + \sin^2\alpha\end{pmatrix} = I$ and $\det[R] = \sin^2\alpha + \cos^2\alpha = +1$.

Part (a) · 1 point

Degrees of freedom and constraints

How many degrees of freedom has the system particle + wedge? Classify the constraint that keeps P on the incline, as seen in S_0 and in S_1.

Step 1. Count

The wedge is a rigid body in the plane (3 coordinates); the table fixes its height and, since it does not tip, its orientation: 1 coordinate X remains. The particle has 2 coordinates in the plane; staying on the incline removes one, leaving s. In total **2 degrees of freedom**, (X, s). Unlike a fixed incline, $X(t)$ is an unknown: the motion of the support depends on the motion of the particle.

Step 2. Classification

The condition "P on the incline" reads $\overrightarrow{AP}\cdot\mathbf{n} = 0$, i.e. $-(x_P - X)\sin\alpha + (y_P - h)\cos\alpha = 0$ with $\overrightarrow{OA} = X\,\mathbf{i} + h\,\mathbf{j}$, a relation between the positions of P and of the wedge: it is **holonomic** and, written in the system coordinates (X, s) (or (X, x_P, y_P)), it contains no explicit time, so it is **scleronomic**. It would be rheonomic for P alone only if the motion $X(t)$ of the wedge were prescribed from outside; here X is an unknown, so the time enters only through the coordinates. In S_1 the plane is fixed. It is **unilateral**: P can leave the plane upwards, so it holds only while $N \geq 0$. Smooth: the reaction is $\mathbf{N} = N\,\mathbf{n}$, with no component along $\mathbf{u}$ (Notes §6.4).

KEY RESULTS

- 2 degrees of freedom, (X, s).
- P on the incline: holonomic, unilateral, smooth; scleronomic as a relation between the coordinates (X, s) of the system (no explicit t); it would be rheonomic for P alone only if $X(t)$ were a prescribed function of time, which it is not here.

Part (b) · 1.5 points

Kinematics of P

Write the position, velocity and acceleration of P relative to S_1 and in S_0, in the basis $\{\mathbf{n}, \mathbf{u}\}$.

Step 1. Relative motion

$\mathbf{u}$ is fixed in S_1, so $\mathbf{r}_{rel} = \overrightarrow{AP} = s\,\mathbf{u}$, $\mathbf{v}_{rel} = \dot{s}\,\mathbf{u}$, $\mathbf{a}_{rel} = \ddot{s}\,\mathbf{u}$. The relative velocity is tangent to the incline.

Step 2. Absolute motion

$\overrightarrow{OA} = X\,\mathbf{i} + h\,\mathbf{j}$ with the height h of the wedge constant, so $\mathbf{v}_A = \dot{X}\,\mathbf{i}$, $\mathbf{a}_A = \ddot{X}\,\mathbf{i}$. Since $\boldsymbol{\omega}_{10} = \mathbf{0}$ (Notes §3.2):

$$\mathbf{v}_P = \dot{X}\,\mathbf{i} + \dot{s}\,\mathbf{u}, \qquad \mathbf{a}_P = \ddot{X}\,\mathbf{i} + \ddot{s}\,\mathbf{u}$$

Substituting $\mathbf{i} = -\cos\alpha\,\mathbf{u} - \sin\alpha\,\mathbf{n}$:

$$\mathbf{a}_P = (\ddot{s} - \ddot{X}\cos\alpha)\,\mathbf{u} - \ddot{X}\sin\alpha\,\mathbf{n}$$

and in Cartesian components $\mathbf{v}_P = (\dot{X} - \dot{s}\cos\alpha)\,\mathbf{i} - \dot{s}\sin\alpha\,\mathbf{j}$.

KEY RESULTS

- $\mathbf{r}_{rel} = s\,\mathbf{u}$, $\mathbf{v}_{rel} = \dot{s}\,\mathbf{u}$, $\mathbf{a}_{rel} = \ddot{s}\,\mathbf{u}$.
- $\mathbf{a}_P = \ddot{X}\,\mathbf{i} + \ddot{s}\,\mathbf{u} = (\ddot{s} - \ddot{X}\cos\alpha)\,\mathbf{u} - \ddot{X}\sin\alpha\,\mathbf{n}$.

Part (c) · 2.5 points

Equations of motion and the reaction

Write the forces on P in S_1, including the inertia force, and on the wedge. Obtain $\ddot{X}$, $\ddot{s}$ and N, and the reaction of the table on the wedge.

Step 1. Forces on P in S_1

Weight $\mathbf{W} = -mg\,\mathbf{j} = mg\sin\alpha\,\mathbf{u} - mg\cos\alpha\,\mathbf{n}$; reaction $\mathbf{N} = N\,\mathbf{n}$; inertia force of the translating frame (Notes §4.9) $\mathbf{F}_{in} = -m\,\mathbf{a}_A = -m\ddot{X}\,\mathbf{i} = m\ddot{X}\cos\alpha\,\mathbf{u} + m\ddot{X}\sin\alpha\,\mathbf{n}$. There is no Coriolis or centrifugal force ($\boldsymbol{\omega}_{10} = \mathbf{0}$).

Step 2. Projections for P

$m\,\ddot{\boldsymbol{s}}\,\mathbf{u} = \mathbf{W} + \mathbf{N} + \mathbf{F}_{in}$:

$$\mathbf{u}: \quad m\ddot{s} = mg\sin\alpha + m\ddot{X}\cos\alpha \implies \ddot{s} = g\sin\alpha + \ddot{X}\cos\alpha$$

$$\mathbf{n}: \quad 0 = -mg\cos\alpha + N + m\ddot{X}\sin\alpha \implies N = m\big(g\cos\alpha - \ddot{X}\sin\alpha\big)$$

The $\mathbf{u}$ equation is free of the reaction (dynamical decoupling, Notes §6.5), but it still contains the unknown $\ddot{X}$.

Step 3. The wedge

Forces on the wedge: its weight $-Mg\,\mathbf{j}$, the table reaction $R\,\mathbf{j}$ (smooth table, vertical) and $-\mathbf{N} = -N\mathbf{n} = N\sin\alpha\,\mathbf{i} - N\cos\alpha\,\mathbf{j}$. Newton for the translating wedge, $M\ddot{\boldsymbol{X}}\,\mathbf{i} = -Mg\,\mathbf{j} + R\,\mathbf{j} - N\mathbf{n}$:

$$\mathbf{i}: \quad M\ddot{X} = N\sin\alpha, \qquad \mathbf{j}: \quad 0 = R - Mg - N\cos\alpha$$

Step 4. Solve

Substituting N of Step 2 in the wedge equation: $M\ddot{X} = m\sin\alpha\,(g\cos\alpha - \ddot{X}\sin\alpha)$, so $\ddot{X}\,(M + m\sin^2\alpha) = m\,g\sin\alpha\cos\alpha$:

$$\ddot{X} = \frac{m\,g\sin\alpha\cos\alpha}{M + m\sin^2\alpha}$$

Then, with $\sin^2\alpha + \cos^2\alpha = 1$:

$$\ddot{s} = g\sin\alpha + \frac{m\,g\sin\alpha\cos^2\alpha}{M + m\sin^2\alpha} = \frac{g\sin\alpha\,(M + m\sin^2\alpha + m\cos^2\alpha)}{M + m\sin^2\alpha} = \frac{(M+m)\,g\sin\alpha}{M + m\sin^2\alpha}$$

$$N = mg\cos\alpha - \frac{m^2 g\sin^2\alpha\cos\alpha}{M + m\sin^2\alpha} = \frac{M\,m\,g\cos\alpha}{M + m\sin^2\alpha}$$

and $R = Mg + N\cos\alpha$. All three accelerations are **constant**. Check with the system: the vertical external forces are $R - (M+m)g = N\cos\alpha - mg$, which equals $m\,\mathbf{a}_P \cdot \mathbf{j} = -m\ddot{s}\sin\alpha$ (the wedge has no vertical acceleration). An independent check with the Lagrange equations in (X, s) gives the same $\ddot{X}$ and $\ddot{s}$.

Step 5. Numbers ($\alpha = 30^\circ$, $M = 2m$)

$M + m\sin^2\alpha = 2m + m/4 = \frac{9}{4}m$: $\ddot{X} = \dfrac{mg\cdot\frac{1}{2}\cdot\frac{\sqrt{3}}{2}}{\frac{9}{4}m} = \dfrac{\sqrt{3}}{9}g = 0.1925\,g$, $\ddot{s} = \dfrac{3m\,g\cdot\frac{1}{2}}{\frac{9}{4}m} = \dfrac{2}{3}g$, $N = \dfrac{2m\cdot m\,g\frac{\sqrt{3}}{2}}{\frac{9}{4}m} = \dfrac{4\sqrt{3}}{9}mg = 0.7698\,mg$ (smaller than $mg\cos 30^\circ = 0.866\,mg$ on a fixed wedge).

KEY RESULTS

- $\ddot{s} = g\sin\alpha + \ddot{X}\cos\alpha$, $N = m(g\cos\alpha - \ddot{X}\sin\alpha)$, $M\ddot{X} = N\sin\alpha$.
- $\ddot{X} = \dfrac{m\,g\sin\alpha\cos\alpha}{M + m\sin^2\alpha}$, $\ddot{s} = \dfrac{(M+m)\,g\sin\alpha}{M + m\sin^2\alpha}$, $N = \dfrac{M\,m\,g\cos\alpha}{M + m\sin^2\alpha}$.
- Table reaction $R = Mg + N\cos\alpha$.

Part (d) · 1.5 points

Limits, the sign of N and the massless wedge

Check that $N > 0$ for every $M > 0$. Study the limits $M \to \infty$ and $M \to 0$. In the second one a student says: "the particle slides down a plane, so it accelerates along the incline with $g\sin\alpha$ times some factor and presses on it". Discuss.

Step 1. Sign of the reaction

$N = \dfrac{Mmg\cos\alpha}{M + m\sin^2\alpha}$: numerator and denominator are positive for $M > 0$ and $0 < \alpha < \pi/2$, so $N > 0$ throughout the motion (the accelerations are constant, so N is too). The unilateral constraint is never violated, and $R = Mg + N\cos\alpha > Mg > 0$: the wedge stays on the table.

Step 2. Heavy wedge

Dividing numerator and denominator by M and letting $M \to \infty$ ($m/M \to 0$): $\ddot{X} \to 0$, $\ddot{s} \to g\sin\alpha$, $N \to mg\cos\alpha$, the results of a fixed incline.

Step 3. Massless wedge (the trap)

Setting $M = 0$: $N \to 0$, $\ddot{X} \to \dfrac{mg\sin\alpha\cos\alpha}{m\sin^2\alpha} = g\cot\alpha$, $\ddot{s} \to \dfrac{mg\sin\alpha}{m\sin^2\alpha} = \dfrac{g}{\sin\alpha}$. The absolute acceleration is

$$\mathbf{a}_P = g\cot\alpha\,\mathbf{i} + \frac{g}{\sin\alpha}\big(-\cos\alpha\,\mathbf{i} - \sin\alpha\,\mathbf{j}\big) = -g\,\mathbf{j}$$

The particle **falls freely**; the incline exerts no force on it. A massless wedge cannot push back ($M\ddot{X} = N\sin\alpha$ with $M = 0$ forces $N = 0$), so it flies away horizontally with $g\cot\alpha$, exactly the motion that keeps the surface under the falling particle. The first half of the student's statement is harmless: $\ddot{s} = g\sin\alpha \cdot \dfrac{M+m}{M+m\sin^2\alpha}$ is indeed $g\sin\alpha$ times a factor, and this factor is ≥ 1 for every M (because $M + m \geq M + m\sin^2\alpha$), growing from 1 ($M \to \infty$) to $1/\sin^2\alpha$ ($M \to 0$), which gives $\ddot{s} \to g/\sin\alpha$. The error is the second half, "the particle presses on the wedge": $N \to 0$, the incline exerts no force and the particle is in free fall, merely accompanied by the surface.

KEY RESULTS

- $N > 0$ for $M > 0$ and $0 < \alpha < \pi/2$: the particle never leaves the incline; $R > Mg > 0$.
- $M \to \infty$: $\ddot{X} \to 0$, $\ddot{s} \to g\sin\alpha$, $N \to mg\cos\alpha$ (fixed incline).
- $M \to 0$: $N \to 0$, $\ddot{X} \to g\cot\alpha$, $\ddot{s} \to g/\sin\alpha$ and $\mathbf{a}_P \to -g\,\mathbf{j}$: **free fall**; the massless wedge is pushed away just fast enough to stay in contact without force.
- The factor $\ddot{s}/(g\sin\alpha) = (M+m)/(M+m\sin^2\alpha) \geq 1$ grows to $1/\sin^2\alpha$ as $M \to 0$; the student's error is "presses on the wedge", since $N \to 0$.

Part (e) · 2 points

First integrals and the speeds after sliding a distance L

Show that the horizontal linear momentum of the system and its mechanical energy are conserved, explaining why the reaction $\mathbf{N}$ does no net work although it works on each body. For $\alpha = 30°$, $M = 2m$, find $\dot{s}$, $\dot{X}$ and $|\mathbf{v}_P|$ when P has slid a distance L, and the work of $\mathbf{N}$ on P and on the wedge.

Step 1. Horizontal momentum

The external forces on the system $\{P, \text{wedge}\}$ are $-mg\,\mathbf{j}$, $-Mg\,\mathbf{j}$ and $R\,\mathbf{j}$: no horizontal component. $\mathbf{N}$ is internal (action-reaction pair). The linear momentum theorem gives $\frac{d}{dt}p_x = 0$, with $p_x = M\dot{X} + m\,\mathbf{v}_P\cdot\mathbf{i} = M\dot{X} + m(\dot{X} - \dot{s}\cos\alpha)$. From rest, $p_x = 0$:

$$\dot{X} = \frac{m\cos\alpha}{M+m}\,\dot{s}$$

Consequently the centre of mass has $\ddot{x}_G = 0$ and moves only vertically.

Step 2. Energy

Powers: the weights are conservative; $R\,\mathbf{j}\cdot\dot{X}\,\mathbf{i} = 0$; the pair $\mathbf{N}$, $-\mathbf{N}$ has total power

$$\mathbf{N}\cdot\mathbf{v}_P + (-\mathbf{N})\cdot\mathbf{v}_A = \mathbf{N}\cdot(\mathbf{v}_P - \mathbf{v}_A) = N\,\mathbf{n}\cdot\dot{s}\,\mathbf{u} = 0$$

because the relative velocity is tangent to the smooth incline (Notes §6.4). So the mechanical energy of the system is constant (Notes §4.5). With the zero of potential energy at $s = 0$, the height of P changes by $-s\sin\alpha$:

$$\tfrac{1}{2}M\dot{X}^2 + \tfrac{1}{2}m\big[(\dot{X} - \dot{s}\cos\alpha)^2 + \dot{s}^2\sin^2\alpha\big] - mg\,s\sin\alpha = 0$$

Step 3. Kinetic energy for $\alpha = 30^\circ$, $M = 2m$

$\dot{X} = \frac{m\cdot\frac{\sqrt{3}}{2}}{3m}\,\dot{s} = \frac{\sqrt{3}}{6}\,\dot{s}$. For P: $\dot{X} - \dot{s}\cos\alpha = \big(\frac{\sqrt{3}}{6} - \frac{3\sqrt{3}}{6}\big)\dot{s} = -\frac{\sqrt{3}}{3}\dot{s}$, whose square is $\frac{1}{3}\dot{s}^2$, and $\dot{s}^2\sin^2\alpha = \frac{1}{4}\dot{s}^2$, so $|\mathbf{v}_P|^2 = \frac{7}{12}\dot{s}^2$ and $\frac{1}{2}m|\mathbf{v}_P|^2 = \frac{7}{24}m\dot{s}^2$. For the wedge: $\dot{X}^2 = \frac{3}{36}\dot{s}^2 = \frac{1}{12}\dot{s}^2$ and $\frac{1}{2}(2m)\frac{1}{12}\dot{s}^2 = \frac{2}{24}m\dot{s}^2$. Total:

$$E_{kin} = \frac{2+7}{24}\,m\dot{s}^2 = \frac{3}{8}\,m\dot{s}^2$$

Step 4. Speeds after sliding L

With $s = L$ and $mgL\sin 30^\circ = \frac{1}{2}mgL$:

$$\tfrac{3}{8}m\dot{s}^2 = \tfrac{1}{2}mgL \implies \dot{s}^2 = \tfrac{4}{3}gL, \quad \dot{s} = 1.1547\sqrt{gL}, \quad \dot{X} = \tfrac{\sqrt{3}}{6}\cdot\tfrac{2}{\sqrt{3}}\sqrt{gL} = \tfrac{1}{3}\sqrt{gL}$$

$$|\mathbf{v}_P|^2 = \tfrac{7}{12}\cdot\tfrac{4}{3}gL = \tfrac{7}{9}gL, \qquad |\mathbf{v}_P| = \tfrac{\sqrt{7}}{3}\sqrt{gL} = 0.8819\sqrt{gL}$$

Check with (c): constant $\ddot{s} = \frac{2}{3}g$ from rest gives $\dot{s}^2 = 2\ddot{s}L = \frac{4}{3}gL$. The time to slide L is $\sqrt{2L/\ddot{s}} = \sqrt{3L/g}$ and the wedge has moved $\Delta X = \frac{\sqrt{3}}{6}L = 0.2887\,L$ towards $+x$.

Step 5. Work of the reaction on each body

Kinetic energy theorem for P alone: $\frac{1}{2}m|\mathbf{v}_P|^2 = W_{weight} + W_{N\to P}$ with $W_{weight} = \frac{1}{2}mgL$: $W_{N\to P} = \frac{7}{18}mgL - \frac{9}{18}mgL = -\frac{1}{9}mgL$. For the wedge: $W_{-N\to wedge} = \frac{1}{2}(2m)\dot{X}^2 = \frac{1}{9}mgL$. The reaction **does** work on each body (the incline moves), but the two works cancel: energy is transferred from P to the wedge.

KEY RESULTS

- $p_x = M\dot{X} + m(\dot{X} - \dot{s}\cos\alpha) = 0 \Rightarrow \dot{X} = \dfrac{m\cos\alpha}{M+m}\,\dot{s}$; $\frac{1}{2}M\dot{X}^2 + \frac{1}{2}m|\mathbf{v}_P|^2 - mgs\sin\alpha = 0$.
- $\alpha = 30^\circ$, $M = 2m$: $E_{kin} = \frac{3}{8}m\dot{s}^2$, $\dot{s}^2 = \frac{4}{3}gL$ ($\dot{s} = 1.1547\sqrt{gL}$), $\dot{X} = \frac{1}{3}\sqrt{gL}$, $|\mathbf{v}_P| = \frac{\sqrt{7}}{3}\sqrt{gL} = 0.8819\sqrt{gL}$.
- Work of $\mathbf{N}$ on P: $-\frac{1}{9}mgL$; on the wedge: $+\frac{1}{9}mgL$; total 0.

Part (f) · 1.5 points

The absolute path of P

Show that P moves along a straight line in S_0 and find the angle β of this line with the horizontal. Compare it with α for $\alpha = 30^\circ$, $M = 2m$.

Step 1. A straight path

$\mathbf{a}_P = \ddot{X}\,\mathbf{i} + \ddot{s}\,\mathbf{u}$ is a constant vector (part (c)). Integrating twice from rest (Barrow): $\mathbf{v}_P = \mathbf{a}_P\,t$ and $\mathbf{r}_P - \mathbf{r}_P(0) = \frac{1}{2}\mathbf{a}_P\,t^2$, so P moves along the fixed straight line through its initial position with direction $\mathbf{a}_P$.

Step 2. Its slope

In Cartesian components $\mathbf{a}_P = (\ddot{X} - \ddot{s}\cos\alpha)\,\mathbf{i} - \ddot{s}\sin\alpha\,\mathbf{j}$; the line descends towards $-x$ with

$$\tan\beta = \frac{\ddot{s}\sin\alpha}{\ddot{s}\cos\alpha - \ddot{X}} = \frac{\sin\alpha}{\cos\alpha - \dfrac{\ddot{X}}{\ddot{s}}}$$

From (c), $\dfrac{\ddot{X}}{\ddot{s}} = \dfrac{m\cos\alpha}{M+m}$ (the same ratio as $\dot{X}/\dot{s}$ in (e)), so $\cos\alpha - \dfrac{m\cos\alpha}{M+m} = \dfrac{M\cos\alpha}{M+m}$ and

$$\tan\beta = \frac{(M+m)\sin\alpha}{M\cos\alpha} = \left(1 + \frac{m}{M}\right)\tan\alpha$$

Step 3. Numbers

$\tan\beta = \frac{3}{2}\cdot\frac{1}{\sqrt{3}} = \frac{\sqrt{3}}{2} = 0.8660$, $\beta = 40.89^\circ$. The particle descends **more steeply** than the incline, because the incline recedes under it. Check with (e): $|v_{Py}|/|v_{Px}| = \sqrt{gL/3}/\frac{2}{3}\sqrt{gL} = \frac{\sqrt{3}}{2}$.

KEY RESULTS

- $\tan\beta = \dfrac{\ddot{s}\sin\alpha}{\ddot{s}\cos\alpha - \ddot{X}} = \left(1+\dfrac{m}{M}\right)\tan\alpha.$
- $\alpha = 30^\circ$, $M = 2m$: $\tan\beta = \frac{\sqrt{3}}{2}$, $\beta = 40.89^\circ > \alpha$.

Interpretation and checks

- **Dimensional check:** $\ddot{X}$ and $\ddot{s}$ are g times a ratio of masses times trigonometric factors (m s^{-2}); $N = Mmg\cos\alpha/(M + m\sin^2\alpha)$ is kg·m s^{-2} = N; $\dot{s}^2 = \frac{4}{3}gL$ is m^{2}s^{-2}; the works are mgL times numbers (J).
- **Limiting cases:** $M \to \infty$ recovers the fixed incline ($\beta \to \alpha$); $M \to 0$ gives free fall ($\beta \to 90^\circ$, $\tan\beta \to \infty$); $\alpha \to 0$ gives $\ddot{X} = \ddot{s} = 0$, $N = mg$ (nothing moves); $\alpha \to \pi/2$ gives $N \to 0$, $\ddot{X} \to 0$, $\ddot{s} \to g$ (the particle falls along the vertical face).
- **Common mistakes in the exam:** (1) using $N = mg\cos\alpha$ as on a fixed incline; (2) forgetting the inertia force $-m\ddot{X}\,\mathbf{i}$ in S_1, or giving it the wrong sign; (3) writing energy conservation for P alone (the reaction works on P: $-\frac{1}{9}mgL$ here); (4) assuming the particle's absolute path is parallel to the incline.